

# A Scale–Range Obstruction to QCD-Only Emergent Gravity

Induced Einstein Terms, Confinement Gaps, and the Limits of Large- $N$  Enhancement

Nektarios Kourakis

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## Abstract

Quantum chromodynamics supplies a concrete example in which strongly coupled vacuum fluctuations induce a local Einstein–Hilbert term, while confinement simultaneously produces a gapped color-singlet spectrum. These two facts are sometimes invoked, separately or together, in discussions of gravity emerging from strong dynamics. We formulate a quantitative obstruction showing that, within a broad single-scale class of four-dimensional confining gauge theories, large- $N$  enhancement of the induced curvature coefficient cannot by itself generate long-range gravity. Under standard planar counting, the induced Einstein coefficient scales as  $C_R \sim N^2 \Lambda^2$ , whereas the lightest physical spin-2 mass scales as  $m_2 \sim \Lambda$  with a nonzero planar limit. Hence increasing  $N$  can parametrically increase the effective Planck scale but leaves the tensor correlation length  $m_2^{-1}$  finite. We sharpen this statement into a spectral-support theorem: large- $N$  enhancement can increase residues and local moments of a gapped stress-tensor correlator, but it cannot move the lower edge of its spin-2 spectral support to zero. We call the resulting statement the *scale–range obstruction*. Using the QCD Adler–Zee estimate and continuum large- $N$  glueball data, we show that matching the observed curvature coefficient by  $N$ -enhancement alone would require an illustrative  $N \sim 5 \times 10^{19}$ , while the tensor range remains of order  $10^{-16}$  m rather than macroscopic. At one astronomical unit the required tensor mass is below  $1.4 \times 10^{-18}$  eV, more than  $10^{27}$  below the QCD tensor-glueball scale. The result does not exclude emergent gravity in general; instead it identifies the minimum structural ingredient that a QCD-triggered model must add: a second infrared scale, an additional gapless singlet sector, non-four-dimensional/holographic dynamics, or a departure from the standard local spectral assumptions. The analysis provides a compact falsification criterion for claims that ordinary QCD confinement alone generates observed four-dimensional gravity.

**Keywords:** emergent gravity; induced gravity; QCD; confinement; large  $N$ ; glueballs; mass gap; Adler–Zee

## 1 Motivation and scope

The observation that quantum matter induces local curvature terms is old and well established. In Sakharov’s induced-gravity program and in the Adler–Zee formulation, integrating quantum fields on a background metric generates an effective action containing an Einstein–Hilbert term [1, 2, 3]. For gluonic QCD, Donoghue and Menezes gave a modern nonperturbative estimate of the induced coefficient using lattice input, perturbative QCD and an operator-product expansion [4].

A logically stronger claim is that the same nongravitational matter theory generates a *physical long-range gravitational degree of freedom*. This distinction has been analyzed explicitly in emergent-gravity constructions based on stress-tensor correlators. In particular, Betzios, Kiritsis and Niarchos emphasized the contrast between induced and emergent gravity, the separate spin-0 and spin-2 spectral sectors, and the fact that a gapped hidden theory generically produces massive rather than massless tensor exchange [5]. Recent work has sharpened the infrared spectral question by focusing on the appearance of zero-momentum spin-2 spectral weight [6].

The present paper therefore does *not* claim novelty for the distinction between induced and emergent gravity, nor for the spectral decomposition itself. Instead we isolate a quantitative obstruction relevant to a narrower hypothesis: can ordinary QCD-like confinement, possibly amplified by large  $N$ , simultaneously account for (i) the weakness of gravity and (ii) its macroscopic range? The answer is negative under a small set of transparent assumptions. The obstruction is useful because it shows why solving the Planck-scale hierarchy through multiplicity is not the same problem as producing a gapless tensor channel.

## 2 Assumptions

Consider a family of four-dimensional, Poincaré-invariant, unitary confining gauge theories labeled by  $N$ . We assume:

- (A1) **Single infrared scale.** In the confining regime, all non-Goldstone color-singlet masses are set by one dynamical scale  $\Lambda$  up to dimensionless  $N$ -dependent factors.
- (A2) **Gapped tensor channel.** The lightest physical state with the quantum numbers sourced by the transverse traceless part of the conserved stress tensor has

$$m_2(N) = a(N)\Lambda, \quad \lim_{N \rightarrow \infty} a(N) = a_\infty > 0. \quad (1)$$

- (A3) **Standard planar counting.** The renormalized coefficient of the induced Einstein term has leading scaling

$$C_R(N) = b(N)N^2\Lambda^2, \quad \lim_{N \rightarrow \infty} b(N) = b_\infty > 0, \quad (2)$$

where  $C_R$  is defined by  $S_{\text{eff}} \supset \int d^4x \sqrt{-g} C_R R$ .

- (A4) **No independent gapless gravitational sector.** Long-range tensor exchange, if present, must arise from the physical stress-tensor sector of the confining theory itself; no fundamental metric propagator, separate conformal sector, extra-dimensional zero mode or additional tuned gapless singlet is inserted by hand.

Assumptions (A1)–(A3) are standard for ordinary large- $N$  Yang–Mills. Assumption (A4) defines the “QCD-only” class being tested.

## 3 Induced curvature and tensor range are independent scales

The background-field effective action can contain

$$S_{\text{eff}}[g] = \int d^4x \sqrt{-g} [-\Lambda_{\text{vac}} + C_R R + O(R^2)]. \quad (3)$$

The Adler–Zee relation expresses the finite matter contribution to  $C_R$  through a suitably renormalized trace correlator [2, 4]. This is a local derivative-expansion coefficient.

By contrast, the physical range of a tensor-mediated interaction is controlled by the nearest nonanalyticity of the transverse spin-2 correlator. For a spectral density with support beginning at  $s = m_2^2 > 0$ , the dispersive contribution is analytic for  $|q^2| < m_2^2$ , and the static exchange is Yukawa suppressed at large separation,

$$V_2(r) \propto -\frac{e^{-m_2 r}}{r} \times (\text{residue and tensor structure}). \quad (4)$$

This general structure follows directly from the stress-tensor spectral representation and is explicit in Ref. [5]. A true  $1/r$  tensor force instead requires spectral support reaching zero in the appropriate

channel (for example an isolated massless pole, or another infrared singular structure capable of generating the same long-distance behavior).

## 4 Spectral-support theorem

Let the connected transverse-traceless stress-tensor two-point function admit, after local subtractions, a Källén–Lehmann representation

$$\Pi_2^{(N)}(Q^2) = P_N(Q^2) + \int_{m_{2,N}^2}^{\infty} ds \frac{\rho_2^{(N)}(s)}{s + Q^2}, \quad \rho_2^{(N)}(s) \geq 0, \quad (5)$$

for Euclidean  $Q^2 \geq 0$ , where  $P_N$  is a subtraction polynomial. Suppose that the spin-2 threshold is uniformly separated from zero in confinement units,

$$m_{2,N} \geq c\Lambda, \quad c > 0, \quad (6)$$

and that large- $N$  changes the normalization of the spectral density but does not introduce a second sector or a new denominator whose zero defines an additional pole.

**Proposition 1 (spectral support cannot be moved by residue enhancement).** *Under Eq. (6), any  $N$ -dependent multiplicative enhancement of  $\rho_2^{(N)}$  changes residues and spectral moments but cannot generate a  $1/Q^2$  singularity at  $Q^2 = 0$ . In particular, if  $\rho_2^{(N)}(s) = N^2 \hat{\rho}_2^{(N)}(s)$  with support contained in  $[c^2\Lambda^2, \infty)$ , then the normalized correlator  $\Pi_2^{(N)}/N^2$  is analytic in the disk  $|Q^2| < c^2\Lambda^2$  up to the subtraction polynomial, for every  $N$  and in any locally uniform large- $N$  limit.*

**Proof.** For  $|Q^2| < c^2\Lambda^2$  one has  $|Q^2/s| < 1$  throughout the spectral support, so

$$\frac{1}{s + Q^2} = \frac{1}{s} \sum_{k=0}^{\infty} \left( -\frac{Q^2}{s} \right)^k. \quad (7)$$

On every compact subdisk  $|Q^2| \leq r < c^2\Lambda^2$ , the series converges uniformly in  $s \geq c^2\Lambda^2$ . After the standard subtractions required for the ultraviolet behavior, term-by-term integration therefore gives an analytic Taylor expansion around  $Q^2 = 0$ . Multiplying the positive measure by  $N^2$  modifies the Taylor coefficients but not the radius set by the closest point of spectral support. Hence residue enhancement alone cannot create a new singularity at the origin. A massless pole can arise only if spectral support itself approaches zero, or if an additional dynamical denominator develops a zero through mixing with another sector.  $\square$

This proposition makes the key statement independent of dimensional analysis. Large  $N$  can enhance a local induced coefficient because such coefficients are spectral moments; it cannot, by itself, change the location of the nearest physical singularity. The second possibility mentioned in the proof is important: coupled hidden/visible constructions can develop DGP-like poles whose mass scales down with  $N$  even when the hidden QFT is gapped [5]. Such a pole is an explicit loophole because it requires an additional sector and an inter-sector coupling, and therefore lies outside assumption (A4).

## 5 Scale–range obstruction theorem

**Proposition 2 (scale–range obstruction).** *Under assumptions (A1)–(A4), increasing  $N$  can parametrically increase the induced curvature scale but cannot parametrically increase the physical tensor correlation length. More precisely,*

$$M_{\text{ind}}(N) \equiv \sqrt{2C_R(N)} \sim N\Lambda, \quad \xi_2(N) \equiv m_2(N)^{-1} \sim \Lambda^{-1}, \quad (8)$$

so that

$$\frac{M_{\text{ind}}}{m_2} = \frac{\sqrt{2b(N)}}{a(N)} N \quad \text{but} \quad \Lambda \xi_2 = \frac{1}{a(N)} \longrightarrow \frac{1}{a_\infty} < \infty. \quad (9)$$

Consequently, a large hierarchy in the induced Einstein coefficient does not imply approach to a long-range tensor phase.

**Proof.** Equation (2) gives  $M_{\text{ind}} = \sqrt{2b(N)} N \Lambda$ . Equation (1) gives  $\xi_2 = [a(N)\Lambda]^{-1}$ . Taking the planar limit yields Eq. (9). The first ratio grows linearly with  $N$ , while the dimensionless correlation length tends to a constant. Therefore large  $N$  can suppress an effective gravitational coupling measured in units of  $\Lambda$ , but does not move the lowest tensor singularity toward  $q^2 = 0$ .  $\square$

The proposition is deliberately modest. It is not a new proof of a mass gap, nor a new derivation of planar counting. Its content is the incompatibility of using the *same* large- $N$  lever to solve both the coupling-strength and range requirements of observed gravity in a single-scale confining theory.

## 6 Continuum Yang–Mills input

Continuum lattice calculations strongly support assumption (A2). Athenodorou and Teper find for pure  $SU(N)$  Yang–Mills in  $3 + 1$  dimensions that the lightest tensor glueball has a smooth nonzero planar limit [7]:

$$\frac{m_{2++}}{\sqrt{\sigma}} \xrightarrow{N \rightarrow \infty} 4.599(14), \quad (10)$$

with leading corrections of order  $1/N^2$ . Thus the physical tensor correlation length remains finite in confinement units even as the number of gluonic degrees of freedom diverges.

The scalar trace sector behaves differently. The connected stress-tensor correlator carries the standard  $N^2$  planar normalization, so an induced local curvature coefficient naturally scales as  $N^2 \Lambda^2$  when the renormalization prescription is held fixed across the family [8]. This is precisely the separation encoded by Proposition 1.

## 7 Numerical QCD corollary

Donoghue and Menezes obtain for gluonic QCD [4]

$$C_R^{(3)} \equiv \frac{1}{16\pi G_{\text{ind}}} = (0.0095 \pm 0.0030) \text{ GeV}^2. \quad (11)$$

For observed gravity, using the reduced Planck mass  $\overline{M}_{\text{Pl}} \simeq 2.435 \times 10^{18} \text{ GeV}$ ,

$$C_R^{\text{obs}} = \frac{1}{16\pi G_N} = \frac{\overline{M}_{\text{Pl}}^2}{2} \simeq 2.96 \times 10^{36} \text{ GeV}^2. \quad (12)$$

Hence

$$\frac{C_R^{\text{obs}}}{C_R^{(3)}} \simeq 3.1 \times 10^{38}. \quad (13)$$

If one uses the simplest planar scaling  $C_R(N)/C_R(3) \simeq (N/3)^2$  merely as an order-of-magnitude exercise, matching Eq. (12) would require

$$\boxed{N_{\text{strength}} \simeq 5.3 \times 10^{19}}. \quad (14)$$

This number should not be interpreted as a viable model prediction: it neglects changes of dynamics, renormalization and possible species/causality constraints. Indeed large- $N$  gauge theories coupled to gravity are independently subject to parametric bounds of order  $N \lesssim M_{\text{Pl}}/\Lambda_{\text{QCD}}$  from causality/species arguments [9, 10]. Equation (14) is used only to expose the scale of the coupling problem.

The range problem is independent and much more severe. Taking a representative QCD tensor mass  $m_2 \sim 2\text{--}2.4$  GeV gives

$$\xi_2 = \frac{\hbar c}{m_2} \sim (0.8\text{--}1.0) \times 10^{-16} \text{ m}. \quad (15)$$

To mediate an unsuppressed force over one astronomical unit  $R_{\text{AU}} = 1.496 \times 10^{11}$  m, a Yukawa mediator must satisfy approximately

$$m_2 \lesssim \frac{\hbar c}{R_{\text{AU}}} \simeq 1.32 \times 10^{-18} \text{ eV}. \quad (16)$$

Thus the QCD tensor scale exceeds the astronomical-range requirement by roughly

$$\boxed{\frac{m_{2,\text{QCD}}}{m_{\text{AU}}} \sim 10^{27}}. \quad (17)$$

Increasing  $N$  does not remove this  $10^{27}$  range mismatch because Eq. (10) remains nonzero at  $N \rightarrow \infty$ .

## 8 Large- $N$ strength matching meets a causality threshold

The illustrative value in Eq. (14) can be compared with an independent parametric threshold derived from higher-spin causality. Kaplan and Kundu find that confining large- $N$  gauge theories weakly coupled to gravity require additional higher-spin/string-like structure once

$$N \gtrsim \frac{M_{\text{Pl}}}{\Lambda_{\text{conf}}}, \quad (18)$$

up to order-one and convention-dependent factors [9, 10]. For a QCD-scale confinement scale  $\Lambda_{\text{conf}} \sim 0.2\text{--}0.3$  GeV, Eq. (18) corresponds parametrically to  $N$  of order  $10^{19}\text{--}10^{20}$  depending on whether reduced or unreduced Planck conventions are used. The strength-matching estimate  $N_{\text{strength}} \simeq 5.3 \times 10^{19}$  therefore lies at or beyond the onset of the regime in which a purely QCD-like description coupled only to weak Einstein gravity is not self-contained: causality demands new higher-spin interactions or a string-like gravitational completion.

This is not a strict numerical exclusion because Eq. (18) is parametric. It is nevertheless structurally important. The same large- $N$  lever invoked to repair the coefficient mismatch drives the theory into a regime where assumption (A4) ceases to be tenable. Thus the “QCD-only” strategy encounters two distinct barriers: the tensor spectral support remains gapped, and the large- $N$  value required to match the observed coefficient is already at the scale where independent causality arguments demand extra degrees of freedom.

## 9 A conditional strength–causality no-overlap criterion

The causality threshold can be combined algebraically with the induced-curvature scaling to obtain a second obstruction that does not involve the tensor range. Write

$$C_R(N) = \kappa N^2 \Lambda^2, \quad C_R^{\text{obs}} = \eta M_*^2, \quad (19)$$

where  $M_*$  is the gravitational scale used in the causality estimate and  $\eta = O(1)$  accounts for Planck-mass convention (for the reduced Planck mass used in Eq. (12),  $\eta = 1/2$ ). Parametrize the onset of the higher-spin/stringy regime as

$$N_c = \zeta \frac{M_*}{\Lambda}, \quad \zeta = O(1). \quad (20)$$

**Proposition 3 (conditional strength–causality no-overlap).** *If the single-sector QCD-like*

description is required to remain below  $N_c$ , then throughout that regime

$$\frac{C_R(N)}{C_R^{\text{obs}}} < \frac{\kappa\zeta^2}{\eta}. \quad (21)$$

Therefore matching the observed Einstein coefficient before the causality/new-physics threshold requires the necessary condition  $\kappa\zeta^2 \gtrsim \eta$ .

**Proof.** Substitute the upper limit  $N < N_c$  from Eq. (20) into Eq. (19). Then  $C_R < \kappa\zeta^2 M_*^2$ , and division by  $\eta M_*^2$  gives Eq. (21).  $\square$

A useful  $SU(3)$  anchor follows directly from Eq. (11). If the confinement scale is identified with the conventional string-tension scale  $\sqrt{\sigma} \simeq 0.44$  GeV, then

$$\kappa_3^{(\sigma)} \equiv \frac{C_R^{(3)}}{9\sigma} \simeq 5.45 \times 10^{-3}. \quad (22)$$

Using instead a lower phenomenological confinement scale  $\Lambda = 0.2\text{--}0.3$  GeV gives the broader estimate

$$\kappa_3 \simeq 1.2 \times 10^{-2}\text{--}2.6 \times 10^{-2}. \quad (23)$$

For  $\eta = 1/2$  and  $\zeta = 1$ , the maximum coefficient attainable before the parametric causality threshold is therefore only about 1% for the string-tension normalization, or about 2–5% for the lower confinement-scale convention. Order-one uncertainty in  $\zeta$  prevents interpreting this as a sharp exclusion. The result is instead a conditional no-overlap statement: with the QCD-sized Adler–Zee coefficient, coefficient matching is not naturally completed inside the same weakly-gravitating, single-sector large- $N$  regime whose extrapolation was invoked to achieve it.

This criterion is complementary to the scale–range obstruction. Proposition 3 concerns the *strength* of the local Einstein term before additional UV/causality structure becomes mandatory, whereas Propositions 1–2 concern the *range* encoded in the infrared spin-2 spectral support. A candidate QCD-origin model must pass both tests.

## 10 A dimensionless falsification plane

For any proposed single-scale strong-dynamics origin of gravity, define two independent dimensionless diagnostics at a target distance  $R$ :

$$\mathcal{S} \equiv \frac{C_R}{C_R^{\text{obs}}}, \quad \mathcal{L}(R) \equiv m_2 R. \quad (24)$$

A model that aims to reproduce observed long-range gravity must reach simultaneously

$$\mathcal{S} \sim 1, \quad \mathcal{L}(R) \ll 1 \quad (25)$$

for all tested macroscopic  $R$ . Ordinary QCD lies in the opposite corner:  $\mathcal{S} \sim 3 \times 10^{-39}$  for the induced coefficient and  $\mathcal{L}(1 \text{ AU}) \sim 10^{27}$  for the tensor gap. Planar enhancement can move the theory horizontally in  $\mathcal{S}$  but, under assumptions (A1)–(A3), does not drive  $\mathcal{L}$  toward zero. This “falsification plane” is a compact way to separate two requirements that are frequently conflated.

## 11 What must a successful QCD-triggered model add?

The obstruction is conditional, so it also classifies the viable escape routes. A model relating QCD dynamics to gravity must violate at least one assumption:

(E1) **Second infrared scale:** generate a parametrically smaller scale  $\mu \ll \Lambda_{\text{QCD}}$  controlling the tensor gap, while QCD controls another sector.

- (E2) **Additional gapless singlet sector:** QCD may trigger or mix with a distinct sector whose spin-2 spectral weight reaches zero. This is no longer QCD-only.
- (E3) **Emergent extra dimension/holography:** a higher-dimensional massless graviton may emerge even though the four-dimensional glueball tower is gapped; such mechanisms lie outside assumption (A4) and are familiar from gauge/gravity duality [5, 11].
- (E4) **Nonlocal or nonstandard infrared dynamics:** abandon the ordinary four-dimensional local Källén–Lehmann picture, or produce a critical continuum rather than a gapped hadronic spectrum.
- (E5) **Fundamental/independent metric sector:** retain QCD as a contributor to the Einstein coefficient but not as the origin of the propagating graviton; this returns to induced rather than matter-only emergent gravity.

This classification turns the original intuition “strong dynamics may have gravity as a side effect” into a sharper research question: *what new structure converts a QCD-scale reorganization into an independent near-zero tensor scale?*

## 12 Audit of minimal escape routes

It is useful to ask whether the obstruction can be evaded with a genuinely minimal addition.

**QCD-triggered ultralight scalar or dilaton.** Approximate scale invariance or an additional singlet can generate a very light  $0^{++}$  mode. Such fields are well studied in walking theories and ultralight-dilaton phenomenology [12, 13]. This can create a second infrared scale, but it does not supply the transverse massless spin-2 structure of general relativity. If coupled appreciably to the QCD trace, it also mediates composition-dependent or scalar fifth forces unless its couplings are highly suppressed. It is therefore an escape from the single-scale assumption, not a QCD derivation of tensor gravity.

**Near-conformal hidden gauge dynamics.** Walking or near-conformal dynamics can generate large hierarchies and unusually light scalar states. However, a parametrically light scalar does not imply that the lightest  $2^{++}$  state becomes parametrically light. A positive model would need an explicit mechanism driving the spin-2 spectral threshold toward zero while preserving unitarity and universal coupling; ordinary near-conformal dynamics does not provide this automatically.

**Coupled hidden sector / DGP-like pole.** A distinct strongly coupled sector connected through stress-tensor interactions can generate an additional pole through the zero of a coupled-sector denominator. Betzios, Kiritsis and Niarchos exhibit precisely this mechanism and find a DGP-like light pole whose mass can scale down with  $N$  [5]. This is a genuine evasion of Proposition 1 because the pole is not obtained by rescaling the spectral weight of a single gapped QCD correlator. It also demonstrates why assumption (A4) is essential: the successful mechanism is no longer QCD-only.

**Massless composite spin-2 in four-dimensional local QFT.** Trying instead to obtain an exactly massless composite spin-2 particle inside an ordinary Lorentz-invariant local QFT encounters the Weinberg–Witten obstruction under its standard assumptions [14]. Holography and extra-dimensional descriptions can evade those assumptions, but again introduce structure beyond a single four-dimensional QCD sector.

This audit did not reveal a minimal positive extension in which ordinary QCD alone both creates the observed Planck coefficient and produces a macroscopic spin-2 range without an additional sector, new infrared scale or non-four-dimensional dynamics. For that reason the scientifically defensible



result of the present analysis is an obstruction theorem and a classification of necessary escape routes, rather than a claimed new theory of gravity.

## 13 Relation to previous work and novelty claim

Several ingredients of the present argument are established. Adler–Zee induced curvature from stress-tensor correlators is standard [2, 3]; QCD’s contribution was computed in Ref. [4]; the distinction between induced and emergent gravity and the spin-0/spin-2 spectral analysis were developed in detail in Ref. [5]; and the nonzero large- $N$  tensor-glueball gap is a lattice result [7]. We therefore make no priority claim for any of those facts.

The proposed contribution is the explicit single-sector *spectral-support obstruction*: residue or multiplicity enhancement of a uniformly gapped spin-2 stress-tensor spectral measure cannot generate a zero-momentum tensor singularity. Proposition 2 translates this spectral statement into the scale–range language for single-scale confining gauge theories. We supplement it with the quantitative QCD corollary, a conditional strength–causality no-overlap criterion, and the falsification plane of Eq. (24). The key statement is that multiplicity can enlarge the coefficient of a local Einstein term without moving the physical tensor singularity toward zero momentum. Thus matching the Planck coefficient by large  $N$  and generating macroscopic range are parametrically independent tasks. A literature search performed for this manuscript did not identify the combined result stated as the spectral-support theorem plus the Adler–Zee/lattice/causality QCD corollary. The underlying ingredients are established, so the novelty is methodological and theorem-level synthesis rather than a new microscopic model. Specialist priority review remains necessary before submission.

## 14 Limitations

The theorem does not exclude emergent gravity generally. It excludes only the class defined by (A1)–(A4). It also does not infer the full nonlinear Einstein equations from a two-point function, address universality of coupling beyond the stress tensor, or solve the cosmological-constant problem. The numerical large- $N$  estimate in Eq. (14) is illustrative and should not be used outside its stated assumptions. Finally, pure Yang–Mills lattice results are used to diagnose the confining tensor sector; full QCD contains light pseudoscalar states but no known massless physical spin-2 state, so the central tensor-gap argument is unchanged.

## 15 Conclusions

QCD provides a real induced-curvature effect but not, by itself, an observed long-range gravitational interaction. The reason is not merely that the induced coefficient is numerically too small. In a single-scale confining gauge theory, the coefficient problem and the range problem transform differently under large  $N$ :  $C_R \sim N^2 \Lambda^2$ , while  $m_2 \sim \Lambda$ . Therefore an arbitrarily large hierarchy in the induced Planck scale can coexist with a strictly finite tensor correlation length.

For QCD, the contrast is extreme. The induced curvature coefficient is roughly  $10^{38}$  below the observed one, while the tensor mass is roughly  $10^{27}$  above what would be needed for astronomical-range Yukawa exchange. Large  $N$  can address only the first mismatch under standard planar counting. Moreover, combining the Adler–Zee normalization with the higher-spin causality threshold gives a conditional no-overlap criterion: for QCD-sized  $\kappa$ , the single-sector theory reaches the parametric onset of additional higher-spin/stringy structure before naturally saturating the observed Einstein coefficient. Any serious proposal in which strong dynamics is causally upstream of gravity must therefore introduce an additional mechanism that creates a second, near-zero tensor scale or changes the dimensional/spectral structure of the theory.

This obstruction converts a broad speculative idea into a falsifiable requirement: a QCD-triggered theory of gravity must explain not only why an Einstein coefficient appears, but independently why a



tensor correlation length diverges.

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